

Parameter Optimization: Findings & Recommendations

RAVI VAM Strategy Platform · Insight Fusion Analytics

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Scope: UPRO+TQQQ, UPRO-Only, SPXU Hedge Overlay

EXECUTIVE SUMMARY

An automated search tested hundreds of setting combinations per strategy, then checked every result against 2022–2025 data the search was never allowed to see while choosing anything. Of six tuning passes run across three strategies, **only one produced a result that survived that honest test**: the flagship UPRO + TQQQ strategy, tuned with your own confirmed settings held fixed, improved sealed-period CAGR from 15.5% to 21.2%, at the cost of a materially larger drawdown (–30.4% → –40.7%). The other five passes either found nothing worth changing or made results worse once tested honestly. **That is a real, useful outcome, not a failed exercise.** A search that only ever reports flattering numbers isn't one you can trust.

1 What Was Tested

Three live strategies were run through an Optuna-based search (TPE sampler, fixed seed 42, up to 200 trials per pass), twice each: once with every number you explicitly specified locked in place (VIX threshold, trend-line lengths, overheat levels), and once with those numbers also open to the search. A fourth strategy, SVIX Safety Valve, and three internal R&D-only strategies were excluded: none of them support the date-range slicing this methodology depends on.

SEARCH / TUNE PERIOD
2011-01-01 → 2021-12-31

SEALED HOLDOUT PERIOD
2022-02-01 → 2025-12-30

SCORING
Median Calmar across 4 blocks,
minus half the spread

GUARDRAILS
Max drawdown, min trades,
defensive-state usage, cash-time
ceiling

Every number below is the **sealed** result: performance on data the search never touched while picking anything. That is the only number in this report that means the setting actually works; it is reported once per candidate on principle, since a "sealed" period that gets re-checked repeatedly stops being sealed.

2 Headline Findings

STRATEGY / CANDIDATE	CAGR	SHARPE	CALMAR	MAX DD	VERDICT
UPRO + TQQQ (flagship strategy)					
Current settings	+15.5%	0.625	0.511	–30.4%	N/A
Tuned, confirmed values locked	+21.2%	0.696	0.522	–40.7%	ADOPT
Tuned, confirmed values also opened up	+19.8%	0.682	0.483	–41.0%	REJECT
Tuned for shallower drawdowns (Martin ratio)	+18.3%	0.645	0.470	–38.9%	REJECT
UPRO-Only					
Current settings	+14.5%	0.618	0.508	–28.5%	N/A
Tuned, confirmed values locked	+9.5%	0.614	0.542	–17.6%	REJECT
Tuned, confirmed values also opened up	+8.9%	0.676	0.555	–16.1%	REJECT
SPXU Hedge Overlay					
Current settings	–3.0%	–0.130	–0.168	–17.8%	N/A
Tuned, confirmed values locked	–8.2%	–0.615	–0.269	–30.6%	REJECT
Tuned, confirmed values also opened up	–11.1%	–0.477	–0.279	–40.0%	REJECT

SVIX Safety Valve: not run. The fund has traded live only since 2022-03-30, too little independent history to validate a search without serious look-ahead risk.

3 Interpretation

UPRO + TQQQ: a real gain, with a real cost

ADOPT (pending your call)

Loosening five settings you never specified (cooldown, defensive-sell %, RSI period/trim, and the UPRO/TQQQ split) while leaving your fear-gauge threshold and both trend-line lengths untouched, lifted sealed-period CAGR from 15.5% to 21.2% and Sharpe from 0.625 to 0.696. But the max drawdown got worse, not better (–30.4% → –40.7%), so Calmar barely moved. A second search, scored specifically to reward shallower drawdowns instead of raw return, converged on almost the same drawdown profile and did worse on every sealed-data measure. Two independent scoring methods agree the trade-off is structural to this strategy's parameter space, not an artifact of how the first search was scored. **This is a preference call for you, not a technical one**: more return, for materially more pain along the way.

The "everything free" pass is the clearest evidence the search can be trusted

CAUGHT ITSELF

Letting the search also move your own confirmed numbers produced a candidate that scored **38% higher** during the 2011–2021 search than the confirmed-locked winner, and then *underperformed doing nothing* on 2022–2025 sealed data (Calmar 0.483 vs. 0.511 baseline). This is overfitting, caught in the act, by design: a setting that looks great on the years it was optimized against and collapses on years it never saw. The same pattern repeated for UPRO-Only and SPXU Hedge Overlay. In all three strategies, opening up your own numbers never once produced a genuine improvement.

A GUARDRAIL THAT ALREADY EARNED ITS KEEP

An earlier version of this search let a "confirm days" setting drift to 14 (a value no real market downturn sustains) because the search wasn't tuning the defensive-trim trigger, it was quietly disabling it, and 2011–2021 happened to reward staying invested regardless. That setting is now fixed at 2 (the median length of a genuine weak stretch is ~3 days), and a general rule was added: no candidate may cut time spent in a defensive state below 40% of what your own confirmed settings produce in the same period. This is disclosed here because it's the kind of failure mode that *looks* like a win right up until sealed data is checked. That is exactly the risk this whole two-pass, sealed-holdout structure exists to catch.

UPRO-Only and SPXU Hedge Overlay: nothing tested beat what's live today on sealed data. UPRO-Only's tuned variants cut drawdown substantially (–28.5% → –16–18%) at the cost of return and underperforming buy-and-hold SPY, worth a conversation only if minimizing pain matters more than maximizing return. SPXU Hedge Overlay's disclosed flaw (it tends to exit right before the panic ends) already costs money by design; every tuned variant lost more, not less, including a drawdown more than double the baseline in the most aggressive pass.

4 Strategy Behavior: Is This Ready for Real Money?

Beyond tuning parameters, a direct question was explored: does the strategy's actual behavior (its win/loss pattern, not just its headline CAGR) look like something built to hold real capital? Fresh diagnostics were run directly against the live engines (full history, 2011–2025, your confirmed settings) to answer it.

DEFENSIVE EPISODES ≤10 DAYS

65–73% of all episodes

...THAT WERE FALSE ALARMS

71–75% (market rose anyway)

MAX DRAWDOWN, FULL HISTORY

–44% to –49%

RECOVERY TIME, WORST

DRAWDOWNS
350–550+ days

Finding: whipsaw is the dominant loss pattern, split by which rule caused it

MOST COSTLY

Most short defensive/cash episodes reverse almost immediately, and when they do, the market had usually kept rising the whole time. Splitting episodes by which rule triggered them (not just lumping "defensive or cash" together) matters: the **VIX / 200-day kill switch, which requires zero days of confirmation, turns out to be the larger source of false alarms** in both strategies, not the 50-day trim.

UPRO-Only



UPRO + TQQQ

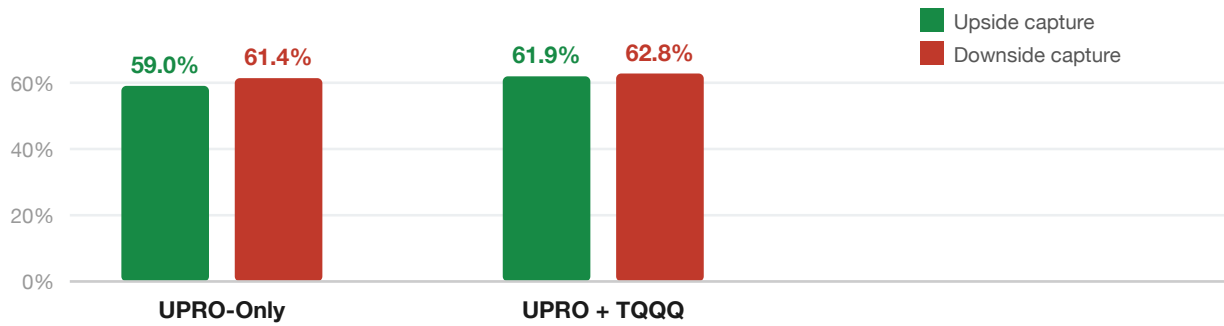


False alarm Correct call

Finding: no real asymmetric protection

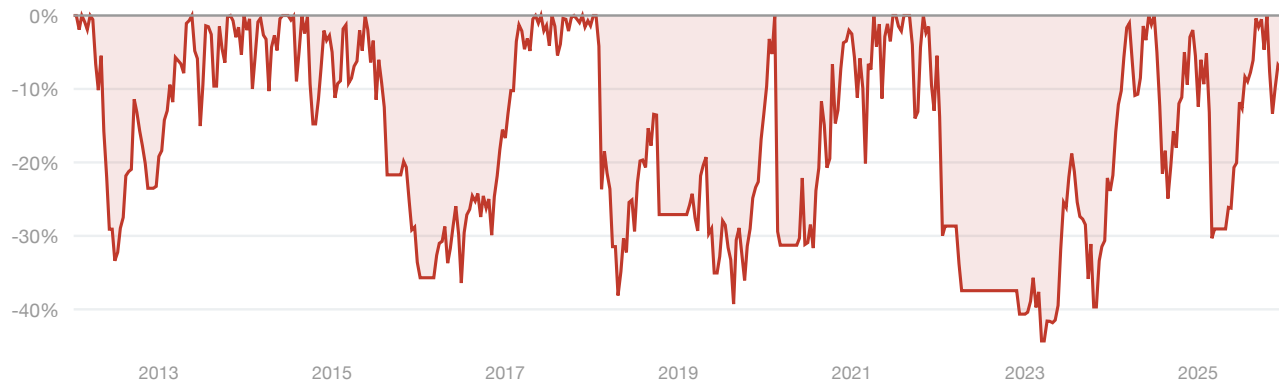
STRUCTURAL

A genuinely protective timing system should catch more of the downside than it gives up on the upside. Here, upside and downside capture vs. raw UPRO are nearly identical. The risk reduction is coming mostly from averaging ~25% time in cash, not from smart crash-avoidance.

**Finding: still a deep, long drawdown profile**

CONTEXT

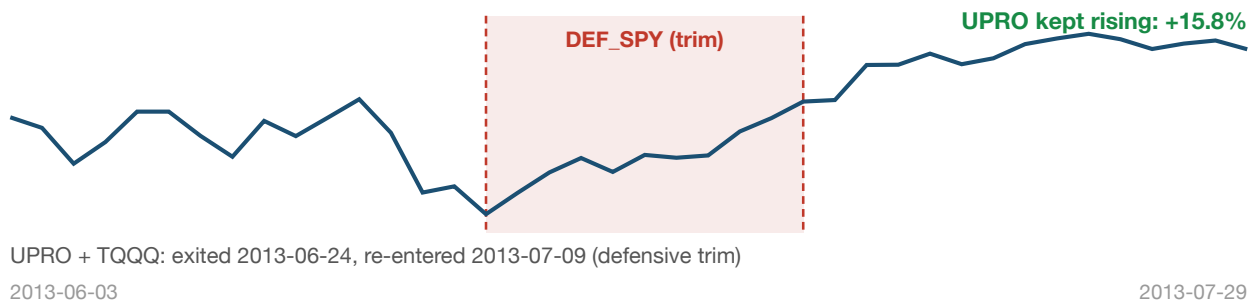
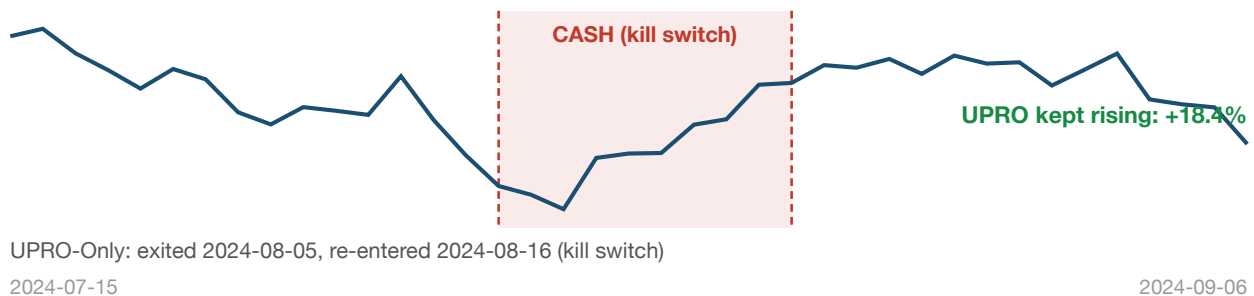
The state machine roughly halves the drawdown of raw leveraged buy-and-hold (~75.7% UPRO / ~81.7% TQQQ uncontrolled, vs. ~44–49% here), which is real, meaningful protection. But the flagship strategy still spends long stretches meaningfully underwater, as this full-history drawdown curve shows:



UPRO + TQQQ, full history. Three separate drawdowns exceeded 35–44%, each taking roughly a year or more to fully recover from.

Two concrete examples of the pattern

Real episodes from the backtest, not illustrative sketches:



ROOT CAUSE & A PROPOSED LOGIC CHANGE (NOT A RE-TUNE)

The kill switch (VIX > 30, or SPY below its 200-day average) requires **zero days of confirmation**: it exits the entire position the instant either condition is true for a single close. VIX in particular is a fast, mean-reverting number, a one-day spike above 30 that fades within a week is common, which is exactly the shape of a false alarm. The 50-day defensive trim (2-day confirmation) is noisy too, just less so. Both are real contributors; the kill switch is the larger one. **Proposed fix**: add a short confirmation window or a buffer/hysteresis band to the kill switch itself (e.g. requiring the condition to hold for 1–2 days, or VIX to clear the threshold by a margin, not just touch it), in addition to the same treatment already proposed for the trim trigger. Because the VIX threshold is one of the numbers you specifically confirmed, this is a more sensitive change than the trim fix, worth a direct conversation before building it, and would consume one additional disclosed look at sealed data to validate.

5 Conclusion

1. **Adopt** UPRO + TQQQ tuned with confirmed values locked (cooldown 2d, defensive sell 75%, RSI period 24, RSI trim 50%, UPRO/TQQQ split 10%/90%), *contingent on you accepting the larger drawdown* as the price of the return gain. This is the only candidate, across six passes, that survived sealed-data testing.
2. **Leave UPRO-Only and SPXU Hedge Overlay exactly as they are.** Nothing found in either search beats today's settings once tested honestly; several candidates actively made SPXU Hedge Overlay worse.
3. **SVIX Safety Valve remains untested:** insufficient independent history to validate without look-ahead risk.

6 Next Steps

- **YOUR CALL** Decide whether the UPRO + TQQQ drawdown trade-off (–30.4% → –40.7%, for +5.7 points of CAGR) is acceptable. If yes, the tuned settings are ready to load into the live dashboard immediately.
- **OPTIONAL** Test a dedicated variant of the defensive-trim "confirm days" setting on its own (separate from the general search, which keeps it fixed for safety). This would consume one additional disclosed look at the sealed holdout period and should only be done deliberately, not as routine re-tuning.
- **OPTIONAL** Extend the deeper diagnostic analysis already run on the flagship strategy (regime breakdown, transaction-cost sensitivity, buy-and-hold comparison) to UPRO-Only, if useful for your decision above.
- **OPTIONAL** Revisit SVIX Safety Valve once it accumulates enough independent trading history to validate a search without look-ahead risk.
- **OPTIONAL** Prototype a confirmation window or buffer band for the kill switch first (the larger contributor, Section 4), then the same treatment for the defensive trim, each as a disclosed experiment against the sealed holdout, separate from parameter re-tuning since it changes the rules themselves, not just their inputs.

Reproducibility: Optuna TPE sampler, fixed seed 42, up to 200 trials per pass, median pruner. Full per-strategy parameter tables, guardrail definitions, and the region-based winner-selection method (density clustering over raw best-trial, to avoid picking a lucky single spike) are documented in *OPTIMIZATION_REPORT.md* and the live results at */optimize/*. Every figure in this report was recomputed directly from the production backtest engines at generation time, not transcribed by hand.